

LAB 3 — ARCHITECTURE SELECTION & WRITTEN JUSTIFICATION

Database Query Performance Profiler

Architecture Selection

We chose **Microservices Architecture** for the Database Query Performance Profiler System.

Architectural Choice and Reasons

1. Independent processing: The system performs separate activities such as slow-query log ingestion, SQL execution-plan analysis, index-candidate recommendation, and weekly digest generation. Independent services allow these activities to be developed and deployed separately.

2. Scalability and fault isolation: Log ingestion and analysis may receive different workloads. Separating them allows the high-volume ingestion and analysis functions to scale independently, while a problem in one service is less likely to stop the complete profiling workflow.

Security Advantage

Service boundaries allow access to be restricted according to responsibility. The ingestion and analysis services can expose only the required APIs, while the profiler data store can be protected from direct access by external users. This supports controlled access to query logs, execution-plan information, and stored recommendations.

Performance Benefit

The ingestion, plan-analysis, and report-generation workloads can operate independently. This reduces the need for a single component to perform every operation sequentially and allows resource-intensive analysis to be scaled separately when the number of slow-query records increases.

Interfaces Shown in the Component Diagram

The diagram contains five core profiler components and shows service/data interactions through Log Ingestion API, Plan Analysis API, Recommendation/Report API, and Report Data API, together with interaction with the database/log source and the DBA/Backend Lead.